SFIC Concept Note

Seva First Innovation Challenge — idea: Hermes Sonar — low-cost, low-power sonar payload for mapping seabed debris [idea no. from portal]

A. Team and idea

Team name / code	Hermes / [team code from portal]
Institute, city, zone	Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad, Telangana · [zone as on portal]
Faculty mentor (if any)	[Name, designation], Department of ECE, CBIT
Level and track	Open · A (product)
Theme and sub-theme	Environment and circular economy · marine waste and clean oceans [match portal wording]
Problem statement (as on portal)	Develop an affordable, low-power sonar payload for autonomous underwater vehicles to map underwater debris and pollution hotspots, enabling circular-economy recovery of marine waste.
Current stage	Idea (architecture and component selection done; signal-chain simulation and RTL in progress)

Team members (2–5):

#	Name	Programme / year	Role and one key skill
1 (lead)	D. Venkat Sai Karthik	B.E. ECE (VLSI Design), 3rd year	Team lead; system architecture and FPGA signal processing in Verilog
2	S. Santhoosh Maadhur	B.E. ECE (VLSI Design), 3rd year	Analog front end and PCB design (chirp TX driver, LNA, ADC)
3	G. Tejas	B.E. ECE (VLSI Design), 3rd year	Embedded firmware and power management (MCU, duty cycling)
4	K. Prabas Reddy	B.E. ECE (VLSI Design), 3rd year	ML debris classifier and mapping software (Python, GIS)
5	Bhaavana	B.E. ECE (VLSI Design), 3rd year	Waterproof housing, test planning, field coordination and benchmarking

B. Summary (≤ 200 words)

About 5.7% of all fishing nets used worldwide are lost at sea every year [1]. In Indian fishing harbours and nearshore grounds, this ghost gear settles on the seabed with tyres and scrap, where it keeps trapping fish and snagging working nets, yet no one maps it because survey-grade sonar costs lakhs of rupees and cameras fail in turbid water. Team Hermes proposes a side-scan sonar payload targeting ₹35,000 per unit and under 5 W average power that mounts on a small boat, ROV or AUV, detects debris on board using FPGA signal processing and a compact neural network, and outputs a GPS-tagged hotspot map labelled by recyclable material. Fishing cooperatives and cleanup NGOs get a “where to dive” map instead of searching blind, recyclers get a predictable supply of nylon net, metal and rubber, and harbour authorities get repeatable data to track cleanup. Our pilot target is to map 1 km² of harbour seabed and help recover at least 200 kg of debris. As five ECE students specialising in VLSI design, our strength is exactly where commercial sonar spends its cost and power: a custom analog front end, Verilog pulse compression on a low-power FPGA, and event-triggered computing.

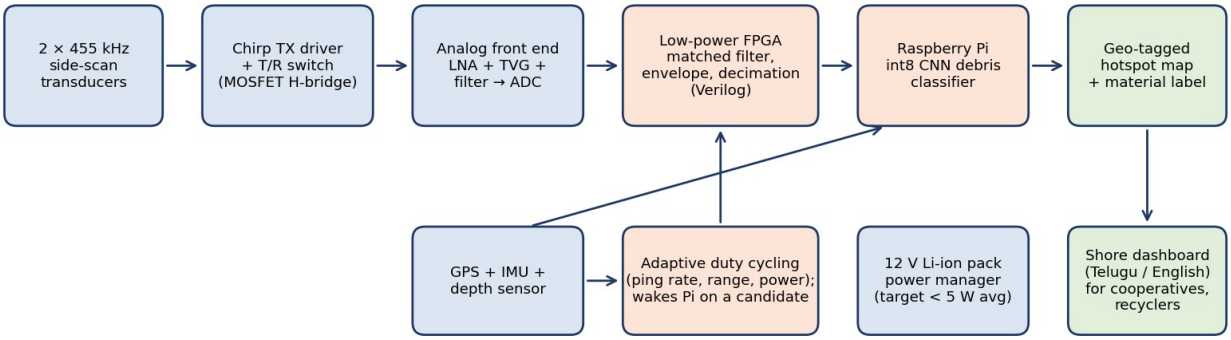
C. Problem (criterion: clarity)

- **The problem, in one sentence:** Fishers and harbour authorities along India’s coast cannot see or locate lost nets, tyres and scrap lying on the seabed of harbours and nearshore grounds, so this debris stays for years, trapping fish, tearing working nets and slowly breaking into microplastics.

- **Scale, with a source:** Globally, about 5.7% of nets, 8.6% of traps and 29% of lines are lost each year [1]; abandoned and lost gear is a major, persistent share of marine litter [2]. India has over 3,000 marine fishing villages and roughly 3.7 million marine fisherfolk [3], all working grounds exposed to this gear.
- **What you saw first-hand:** Not yet done, stated honestly. Before Stage 2 we plan to interview at least 15 fishers and one cleanup NGO at Visakhapatnam Fishing Harbour about net snags, losses and where debris collects, and to survey Hussain Sagar (Hyderabad) shoreline debris. Until then, this note relies on cited reports.
- **Why it persists and what exists today:** Survey-grade side-scan sonar and AUVs cost several lakh rupees and need trained operators, so seabed surveys are rare. Coastal drives such as Swachh Sagar, Surakshit Sagar [5] focus on beaches. Divers search blind, cameras fail in turbid harbour water, and where sonar data exists it is raw imagery that cooperatives and recyclers cannot use.

D. Solution (criteria: originality, feasibility)

- **What it is:** A compact side-scan sonar payload that bolts onto a small boat, ROV or AUV. It detects seabed debris on board and produces a GPS-tagged hotspot map, with each hotspot labelled by likely recyclable material (nylon net, metal, rubber).
- **How it works:** Two 455 kHz transducers send chirp pulses from a MOSFET driver. Echoes pass through our analog front end (LNA, time-varying gain, filter) into an ADC. A low-power FPGA runs matched-filter pulse compression, envelope detection and decimation in Verilog. A Raspberry Pi runs an int8 CNN to classify debris, fuses GPS, IMU and depth data, and builds the map. Adaptive duty cycling lowers ping rate and power over clear seabed and wakes the Pi only when the FPGA flags a candidate object.



Blue: sensing and power Orange: our VLSI/embedded core Green: user output

- **What is new:** (1) Hardware pulse compression on a low-power FPGA plus event-triggered computing, targeting under 5 W average against tens of watts for survey-grade systems. (2) A classifier trained for debris classes, not general seabed imaging. (3) Output tied to recovery: each hotspot carries a material label for recyclers. (4) Open, repairable design at roughly 1/30th the cost of survey sonar.
- **Prototype / pilot status:** Completed: system architecture and component selection. In progress: Python simulation of chirp pulse compression and Verilog RTL for the matched filter. Nothing tested in water yet. Next build step: bench prototype and tank tests at CBIT with planted tyres, nets and metal. Photo/video: to be added. **TRL Level: 2 (target TRL 4 by Dec 2026)**

E. Benchmarking (criteria: originality, cost)

Parameter	Ours (M/T)	Best alternative: survey-grade side-scan towfish	Low-cost: consumer side-imaging fish finder
Debris detection	≥ 80% recall on nets, tyres, metal ≥ 0.5 m; auto-labelled (T)	High-resolution imagery; detection by trained analyst	Imagery on screen only; no detection

Parameter	Ours (M/T)	Best alternative: survey-grade side-scan towfish	Low-cost: consumer side-imaging fish finder
Processing time	Detection on board within ~2 s per swath; map ready at end of survey (T)	Post-processing, hours to days	Real time on screen; no map output
Power and connectivity	< 5 W average, 12 V battery, works offline (T)	Tens of watts, vessel power and cable	~10 W from boat battery
Unit cost (₹)	~35,000 at 50 units (T)	~10–40 lakh (approx.)	~40,000–1,00,000 (approx.)
Yearly running cost (₹)	~5,000: seals, battery (T)	Lakhs: maintenance contract plus operator	Low
Who can use it	Fishers or NGO volunteers after 1-day training; Telugu/English dashboard (T)	Trained hydrographic surveyor	Fishers, but no debris or mapping workflow

One line on the trade-off: We give up range and resolution (about 30–50 m per side versus 100 m+), which is enough for harbours and nearshore grounds where debris concentrates, in exchange for a much lower cost, low power and a ready-to-use hotspot map.

F. Plan and feasibility (criterion: feasibility)

- **Hardest risk and how you have reduced it:** Reliably classifying nets and plastics in sonar images with little labelled data. We start with acoustically strong targets (tyres, metal, bundled nets), build our own labelled set by planting objects in a tank and lake, use public sonar debris data [4], report a confidence score, and send low-confidence detections for diver or camera checks. A partner cooperative or NGO operates the payload from its boat; our team maintains it during the pilot. Approvals: Fisheries Department / port permission for harbour surveys and HMDA permission for Hussain Sagar.

Timeline:

Phase	Activities	Output / milestone
1. Refine prototype (Nov–Dec 2026)	Build AFE and FPGA boards, firmware and housing; tank tests at CBIT with planted objects; measure power	TRL 4 bench prototype; ≥ 200 labelled sonar images; measured power and detection rate
2. Field pilot (Jan–Apr 2027)	Lake trials at Hussain Sagar, then a harbour survey at Visakhapatnam with a cooperative or NGO; diver ground truth	Hotspot map of 1 km²; accuracy vs ground truth; ≥ 200 kg debris recovered and weighed
3. Scale to first adopter (May–Dec 2027)	Custom PCB revision, 5 units built locally; train operators; tie up with a net recycler	5 units in use with first adopter; recycler purchase agreement

G. Cost (criterion: cost)

Prototype / pilot budget:

Item	Qty	Unit cost (₹)	Total (₹)
455 kHz transducers	2	6,000	12,000
AFE and chirp TX driver components	1 set	8,000	8,000
FPGA development board (ECP5 class)	1	12,000	12,000

Item	Qty	Unit cost (₹)	Total (₹)
Raspberry Pi 4 (4 GB) with SD card	1	6,000	6,000
GPS, IMU, depth/pressure sensor	1 set	5,000	5,000
Li-ion battery pack and power board	1	4,000	4,000
Waterproof housing, O-rings, penetrators	1	8,000	8,000
PCB fabrication and assembly	2 rounds	3,000	6,000
3D printing / machining of mounts	1	4,000	4,000
Software (Python, Yosys/nextpnr, QGIS: open source)	–	0	0
Boat hire for field tests	3 days	3,000	9,000
Travel to Visakhapatnam pilot	1 trip	15,000	15,000
Test objects and consumables	–	3,000	3,000
Total			**92,000**

At deployment: cost per unit ~₹35,000 (T, custom PCB at 50 units); cost per beneficiary ~₹120 per fisher household (one unit plus ₹5,000/year upkeep over 5 years = ₹60,000, serving one harbour of ~500 households); who pays: CSR (environmental sustainability), State Fisheries Department harbour upkeep aligned with PMMSY [6], NGOs, with recyclers paying per kg of recovered net.

Cost basis: approximate online prices checked October 2026; quotations to be attached on the portal.

H. Impact (criterion: impact)

Who benefits	Number in pilot	Number at scale	How they benefit
Direct: fisher households at surveyed sites	~200	~20,000 (40 harbours × 500)	Fewer net snags and gear losses; share of recovered-net income
Direct: cleanup teams and recyclers	2 organisations	~25 organisations	Divers know where to search; steady supply of recyclable material
Indirect: coastal communities and marine life	1 harbour	40 harbours	Less ghost fishing, cleaner grounds, data for harbour authorities

Main outcome: kg of seabed debris located and recovered per survey day. Baseline today: near zero at target sites (only beach cleanups). Pilot target: 1 km² mapped, ≥ 20 hotspots, ≥ 80% confirmed by divers, ≥ 200 kg recovered. Measured by GPS logs, diver checks and weighed recovery. All pilot numbers are estimates (T) until measured.

Wider effects and any harm: Less ghost fishing, recyclable nylon and metal recovered, and local repair and assembly skills. Risks: e-waste from batteries and boards, limited by a modular, repairable design and take-back; acoustic disturbance, limited by low power, short surveys and a 455 kHz frequency above most marine mammal hearing; exclusion of non-smartphone users, addressed with printed maps and an offline Telugu/English dashboard.

I. Scale and sustainability (criterion: scalability)

- **Who adopts it:** State Fisheries Departments (Andhra Pradesh coastal harbours, Telangana reservoirs), port authorities, municipal lake bodies such as HMDA, and coastal cleanup NGOs, plugging into PMMSY [6], Swachh Sagar, Surakshit Sagar [5] and CSR funding.
- **How it keeps running after the prize money:** Survey-as-a-service by a team spin-off or trained cooperative operators, charged per km²; buy-back income from recyclers for recovered nets; local PCB assembly in Hyderabad; open documentation so units can be repaired locally.

- **What changes in another state or terrain, and what stays the same:** Changes: turbidity, salinity and seabed type (classifier is retrained on local data), dashboard language, local permits. Same: hardware, signal chain, mapping pipeline and recovery model. Works in rivers, lakes and reservoirs too.

J.Any other information like GIT hub, Patents, Awards etc.

GitHub: [repository link, to be published]. Patents and awards: none yet; we will assess a design patent for the low-power FPGA sonar signal chain after the pilot.

J. Risks, Mitigation Plan, references and declaration

Top risks: (1) Technical, M: low classifier accuracy on nets and plastics; mitigated by strong-target classes first, a planted-object dataset and confidence scores. (2) Technical, M: housing leaks or biofouling; mitigated by pressure testing to 2× working depth, rated penetrators and a 30 m depth limit. (3) Approval, M: delays in harbour or lake permissions; mitigated by starting in the CBIT tank, applying early, and partnering with an NGO that already holds access.

References:

[1] Richardson K., Hardesty B.D., Wilcox C. (2019). Estimates of fishing gear loss rates at a global scale: a literature review and meta-analysis. Fish and Fisheries, 20(6).

[2] Macfadyen G., Huntington T., Cappell R. (2009). Abandoned, lost or otherwise discarded fishing gear. UNEP Regional Seas Reports and Studies 185; FAO Fisheries and Aquaculture Technical Paper 523.

[3] ICAR-CMFRI. Marine Fisheries Census 2016 – India. Central Marine Fisheries Research Institute, Kochi.

[4] Valdenegro-Toro M. (2019). Deep Neural Networks for Marine Debris Detection in Sonar Images. PhD thesis, Heriot-Watt University.

[5] Ministry of Earth Sciences, Government of India (2022). Swachh Sagar, Surakshit Sagar coastal cleanup campaign.

[6] Department of Fisheries, Government of India. Pradhan Mantri Matsya Sampada Yojana (PMMSY): operational guidelines.

Annexures on portal: prototype photos, test data, support letters, demo video link.

Video max 50 MB

Declaration. We declare that this work is original to our team, that all sources are acknowledged, and that the information given is true to the best of our knowledge. (Optional)

Name	Role	Signature / date
D. Venkat Sai Karthik	Team lead	
[Faculty mentor name]	Faculty mentor	
[Coordinator name]	Institute coordinator	